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Professors Paris Avgeriou and David Shepherd
Co-Editors-in-Chief
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Dear Professors Avgeriou and Shepherd,

I am pleased to submit the manuscript “*Post-Execution Defect Attribution on Real-World Defects: An Empirical Comparison of Four Methods on Defects4J and BugsInPy*” for consideration at the Journal of Systems and Software.

The paper compares four post-execution defect-attribution methods—a pre-execution classifier baseline, a failure-only flat prior, a hand-crafted rule-based fusion, and a machine-learned gradient-boosting fusion—on 1,323 real defects assembled from Defects4J (824 Java bugs across 16 projects) and BugsInPy (499 Python bugs across 17 projects), for a total of 6,988 (event, file) prediction rows across 33 projects. To eliminate in-project training-test leakage the evaluation uses a strict Leave-One-Project-Out cross-validation protocol: for every held-out project the classifiers are trained on the other 32 projects and scored only on the target project. Precision@1/3, Recall@5, Mean Reciprocal Rank, simulated triage seconds, and 10-bin Expected Calibration Error are reported for all four methods on the full corpus.

The empirical finding I would flag for the editors is deliberately honest: on real defects under Leave-One-Project-Out, the machine-learned fusion does *not* outperform the pre-execution baseline. The hand-crafted rule-based fusion retains a marginal 0.30-percentage-point Precision@1 lead over the baseline, and the machine-learned fusion is 2.95 pp *worse*. What the study does establish positively is that cross-project pre-execution defect priors on real corpora are extraordinarily well-calibrated (Expected Calibration Error 0.0209), and that calibration—not any ranking-quality improvement from fusion—is the operationally meaningful deployment property. The paper is framed around this reversal-and-calibration finding rather than around a new state-of-the-art claim.

For full transparency I disclose that an earlier version of this line of work used a synthetic failure-event corpus and was declined at three venues. Reviewers correctly identified the synthetic dependency as the principal external-validity weakness. The present manuscript is a substantively different empirical study: the corpus is entirely real Defects4J and BugsInPy defects, the evaluation protocol is strict Leave-One-Project-Out, the headline finding is the reversal of the fused-ML claim rather than its confirmation, and the calibration-first framing is drawn from the real-data results themselves. I am submitting the manuscript to the Journal of Systems and Software as a fresh first-look submission on the strength of these differences.

The manuscript is not under consideration elsewhere, contains no material previously published in a peer-reviewed venue, and reports the author’s sole work. A reproducibility bundle—LOPO event corpus, method implementations, per-event metrics, and calibration-figure scripts—will be

archived on Zenodo with a DOI added at acceptance. I confirm that a large language model was used only for drafting assistance and prose editing; all experimental design, corpus construction, code, results, and interpretation are the author's own.

Thank you for your consideration.

Sincerely,

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